

Pratik Chatse

Sr. Software Engineer (Computer Vision, ML, Deep Learning, Reinforcement Learning)

Bengaluru, India · GitHub · Blog

Summary

Senior Software Engineer with over 5 years of experience specializing in Machine Learning, Computer Vision, and Deep Learning. Proven expertise in developing and deploying production-level AI solutions on cloud and edge devices, particularly within the medical domain. Adept at model optimization, automation, and working in fast-paced, agile environments. Holds an M.Tech in Medical Imaging from IIT Kharagpur.

Skills

- **ML Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
- **Languages:** Python, Elementary C, Bash scripting (novice)
- **Cloud & DevOps:** AWS (SageMaker), GKE, Azure, Docker, CI/CD, Bitbucket
- **Tools & Technologies:** Git, Linux, TensorRT, Raspberry Pi, Arduino, IoT, MATLAB
- **Expertise:** Computer Vision, Deep Learning, Reinforcement Learning, Model Optimization, MLOps, Edge Computing, Data Science, Image Processing

Professional Experience

Sr. Software Engineer (Machine Learning)

Feb 2019 – Feb 2025 (6 years) · Bengaluru, India

- Engineered and deployed real-time AI inference models on edge devices (Raspberry Pi) for live analysis of medical videos, providing AI-assisted annotations during colonoscopy procedures.
- Developed and maintained production-level Python code within a CI/CD pipeline using Jenkins and Bitbucket.
- Utilized AWS SageMaker and GPU instances for cloud-based model training, evaluation, and automation, focusing on model drift and performance monitoring. Optimized model inference latency using NVIDIA TensorRT containers and containerized model serving for edge deployment.
- Led data science experimentation, data curation, and root-cause analysis to improve model accuracy and resolve non-convergence issues.
- Implemented solutions for fine-grained object detection with limited data and upgraded models to the latest State-of-the-Art (SOTA) architectures for performance improvements.
- Developed dashboards for data visualization and KPI tracking, and created high-level design documents for new product features.
- Mentored and managed junior engineers and interns, leading a geographically distributed team.

Embedded IoT Engineer

2018 (2 months) · Singapore

- Prototyped IoT solutions for a marine chemical logistics platform using Raspberry Pi and Arduino.
- Integrated various IoT sensors and devices, working with common communication protocols.
- Contributed to a SaaS platform providing dashboarding and navigation for ship oil tankers.

Projects

Standalone Digital Microscopy & CAD Device

M.Tech Project · June 2017 – June 2018

Developed a low-cost, standalone diagnostic device using Raspberry Pi for rapid cellular and hematological imaging. Implemented on-board image processing and machine learning for point-of-care diagnostics.

Technologies: Raspberry Pi, Arduino, Machine Learning, Deep Learning, IoT, Motor Control.

Motor Imagery Classification for Brain-Computer Interface

B.Tech Project

Implemented a system to classify EEG brain signals for a binary man-machine interface.

Technologies: MATLAB, Biomedical Signal Processing.

Education

M.Tech, Medical Imaging and Informatics

Indian Institute of Technology (IIT), Kharagpur · 2016 – 2018

B.E., Electronics and Communication Engineering

Birla Institute of Technology (BIT), Mesra · 2008 – 2012